

Northern University Bangladesh

Department of CSE

PROJECT PROPOSAL

Otithi

A Homestay Platform for Remote-Area Tourism in Bangladesh

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1 Executive Summary

Otithi is a web-based homestay platform connecting domestic travellers with local families who have rooms available in remote villages and hill areas of Bangladesh. It addresses a gap in accommodation discovery: travellers often cannot find or book a stay in these areas, while local families have unused rooms that could provide income. Travellers can browse by district and village, review practical room and household details, and submit booking requests. Hosts are verified, and listings are checked by an administrator before publication. Accepted requests receive a booking code; guests pay hosts directly in cash, so the platform does not process payments. By using landmark-based directions, transparent condition disclosures, village-level browsing, and verification on both sides, Otithi is designed around the practical needs of rural hosts and travellers.

2 Introduction and Problem Statement

Accommodation platforms often assume that destinations have hotels, street addresses, reliable infrastructure, and hosts able to accept online payments. These assumptions do not hold in many remote parts of Bangladesh. Travellers may be unable to confirm a place to stay, even where families have spare rooms. At the same time, hosts lack a practical way to make those rooms discoverable and manage requests. Otithi proposes a locally adapted platform that makes rural homestays easier to find and book while clearly communicating conditions before a guest travels.

3 Objectives

1. Enable travellers to discover and request homestays by district, village, dates, price, guest count, and relevant room conditions.
2. Help rural hosts publish room details, availability, landmark directions, household information, and house rules.
3. Build trust through administrator approval, guest verification, transparent disclosures, booking codes, and reviews by both hosts and guests.
4. Support village-based group travel and provide local families with an opportunity to earn income from unused rooms.

4 Proposed System and Work Plan

Hosts register and submit identity documents for administrator approval. Approved hosts can list rooms with photos, price, capacity, availability, house rules, and directions based on nearby landmarks and bazaars rather than a formal street address. Before publication, each listing records a standard condition checklist covering toilet type, electricity, water source, mobile-network availability, bedding, and road access. Listings also disclose household and safety details, such as whether the family lives on site, whether there is a separate entrance, and suitability for solo female travellers or families with children.

Travellers browse listings by district and village and filter by dates, number of guests, price, and room conditions. Listings in the same village are presented together, allowing larger groups to

request rooms across neighbouring households. Verified travellers can send booking requests; hosts may review the guest's history and accept or decline. When a request is accepted, Otithi issues a booking code. The guest presents it on arrival and pays the host directly in cash. After the stay, both parties can leave reviews, including feedback on whether the disclosed room conditions were accurate.

The administrator manages host and listing approvals, location records, and reports. The platform is limited to domestic travel within Bangladesh and does not process online payments. Room information is supplied by hosts and reviewed by the administrator before publication.

5 Technology Stack

- **Frontend:** React.js with Tailwind CSS.
- **Backend:** Node.js with Express.js.
- **Database:** MongoDB hosted on MongoDB Atlas.

6 Expected Outcomes and Significance

The expected outcome is a web platform through which travellers can find rural accommodation with clearer expectations, and hosts can manage listings and booking requests. Landmark-based directions and village grouping reflect how travel and accommodation work in remote communities. Two-sided verification, safety information, and mutual reviews aim to support trust between people sharing a family home. The platform may also help local households earn income from otherwise unused rooms and make remote destinations more accessible to domestic tourism.